Tutorial problems on LPC2148 hardware interfacing

1) Do literature survey, and consult the LPC2148 datasheets to design an interface to control a relay using LPC 2148. Choose GPIO port0 pin 10 to control. You may use ULN 2803 relay driver. Draw the circuit diagram connecting relevant pins of LPC 2148, ULN2803 and the relay. Write the C program that can keep switching the relay between ON and OFFstate with a delay 5 0 msec between the switching.

2) Do literature survey, and consult the LPC2148 datasheets to design a module that generates a waveform using LPC 2148. Use the onchip DAC of LPC2148. Generate the waveform and make it available on the AOUT pin. THe waveform is periodic that raises linearly from 0V to VREF, remains in VREF for 100 msec and drops to 0V in a period.

3) Write a C program to display the message **"Hello World"** using Internal UART0 of LPC 2148 at a **baud rate of 3600 bps.** Configure UART0 pins P0.0 (TxD0) and P0.1 (RxD0) through the PINSEL0 register. Set the frame format to 8 data bits, no parity, and 1 stop bit (8-N-1). Assume that the peripheral clock (PCLK) is 20 MHz. Using **Table 167: Baudrates available when using 20 MHz peripheral clock (PCLK = 20 MHz)** in the LPC214x User Manual, determine and program the correct values for U0DLL, U0DLM, MULVAL, and DIVADDVAL to achieve **3600 bps** with minimal error.